A STRUCTURAL SIEVE FOR BERTRAND'S POSTULATE: THE DETERMINISTIC REACH OF THE PRIME BASE AND ITS NEWTON CLOSURE

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Abstract. We take the prime base, not the interval, as the object of study, and measure the deterministic reach of its generative sieve. The Bertrand–Chebyshev theorem is then not a goal but a consequence: asking how far the base reaches collapses the postulate to a single positivity statement about an exact window count — the window itself derived from the base — which we close in the natural generality of the construction.

Starting from a complete prime base $\mathcal{B} = \{q \text{ prime} : q < P_{\max}\}$ we separate two determinisms of the generative sieve. The base is *self-contained* exactly up to $P_{\min} \cdot P_{\max}$, where $P_{\min}$ is the smallest prime that participates in the sieve: every composite in the window $(P_{\max}, P_{\min}P_{\max}]$ is built entirely from the base, and immediately beyond the edge the first composite requiring a prime *outside* the base appears, namely $P_{\min} \cdot \mathrm{nextprime}(P_{\max})$. Its *generative decision* — a survivor is void $\Leftrightarrow$ prime — reaches further, throughout the zone $(P_{\max}, P_{\max}^2)$, and only there does it break. Inside the self-contained window this decision forces every survivor to be a new prime, so the base cannot cover its own window and a prime outside it is necessary: this forcing is the *atom*. The self-contained edge is thus *derived* from the base, not assumed. This yields a generalized window family $(P_{\max}, P_{\min}P_{\max}]$ obtained by *truncated arithmetic* (sieving with a reduced base); the classical Bertrand window $(n, 2n]$ is the *tightest* member, $P_{\min} = 2$, and every wider member follows from it by monotonicity. The whole construction is invariant under dilation of scale: the pair (window, base) is a single self-similar object, and this self-similarity is what the window count inherits.

The reduction sends non-emptiness of the window to a single positivity statement (the *atom*), which the proof settles in two regimes of different character. In the *constructive* regime the mechanism is generative: at every concrete scale it decides the window outright — the sieve is finite and its evaluation decidable, so no counterexample can arise at any fixed step — and there the theorem is closed by structure alone. What a generative process cannot produce is a single finite object for the completed infinite, the uniform statement over all scales at once; that statement is settled not by construction but existentially. It is exactly here that the central binomial coefficient $\binom{2n}{n}$ supplies closure. It enters not as an external estimate but as the object whose prime factorization records the window's new sources: a source $q \in (n, 2n]$ divides $\binom{2n}{n}$ exactly once. It is present not for any fixed window but because the proof must quantify over all windows at once, and it is reached from the paper's own purity law, not borrowed. The extremal window ($P_{\min} = 2$) is then closed by comparing the growth bound $4^n \le (2n+1)\binom{2n}{n}$ with the old-source upper bound; both bounds are reproved within the development, so its main theorem imports no Mathlib Bertrand theorem. These are the same two inequalities Erdős used in 1932: reaching the same object by a different route is a convergence, with historical priority but not exclusivity. The entire development is machine-checked in Lean 4 / Mathlib with no `sorry` and no appeal to Mathlib's proof of Bertrand's postulate.

1. Introduction

Bertrand's postulate asserts that for every integer $n > 1$ there is a prime p with $n < p \le 2n$. Bertrand conjectured it in 1845 on numerical evidence [1]; Chebyshev proved it in 1852 by analytic bounds on ϑ and ψ [2]; Ramanujan gave a short Γ-function argument in 1919 [3]; and Erdős, in 1932, gave the elementary proof that has since become canonical, extracting the statement from divisibility properties of the central binomial coefficient $\binom{2n}{n}$ [4]. Nair

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later observed that the same bounds follow from $\operatorname{lcm}(1,\ldots,2n)$ [5]. Each of these proofs stands on its predecessors; none discards the tools it inherits.

The present work stands on the same lineage, and is explicit about where. Its contribution is not a new inequality but a new *architecture*: we show that Bertrand's postulate is the tightest instance of a self-similar family of window statements, that the window and its right endpoint are *forced by the prime base* rather than posited, and that the entire theorem localizes to a single positivity statement about an exact count. What remains at that one point is quantitative, and there we insert the lightest classical certificate — which turns out to be Erdős's pair of inequalities.

Three things are new here. First, the window $(P_{\max}, 2P_{\max}]$ and, more generally, $(P_{\max}, P_{\min}P_{\max}]$, together with the right endpoint $P_{\min}P_{\max}$, are *derived* from the base $\mathcal{B} = \{q \text{ prime} : q < P_{\max}\}$: the endpoint is exactly the buildability edge of the base, and self-containment provably fails one step beyond it, while the generative decision (void $\Leftrightarrow$ prime) reaches to the square of the anchor (§2, §3). Second, the statements assemble into a generalized family indexed by the smallest participating prime $P_{\min}$, in which Bertrand is the tightest ($P_{\min} = 2$) and every wider member is a corollary by monotonicity (§4); the family is obtained by *truncated arithmetic*, i.e. sieving with a reduced base, and is manifestly invariant under dilation of scale (§5). Third, non-emptiness of the window collapses to a single exact-count statement (the atom), and entire regimes of it close with no counting at all (§6); the residual case is closed on the central binomial coefficient, whose prime factorization records the window's content (§7, §8).

We are careful about provenance. The development is neither independent of Erdős nor a disguised copy of him. The structural reduction is independent and shows *why* the postulate collapses to one atom; the object on which the atom is finally settled — the central binomial — is reached from the internal purity law of the construction, and would be reached here regardless of the historical record. That Erdős settled the same object in 1932 is a convergence on one mathematical object; he holds the historical priority of use, but no exclusivity over it. Every claim below is machine-checked in Lean 4 with Mathlib; the closing certificate is the sole place a Mathlib import about $\binom{2n}{n}$ is used, and Mathlib's own proof of Bertrand's postulate is never invoked.

Origin of the construction. The construction did not begin from Bertrand's postulate, and it is worth recording the observation from which it did begin, since that observation fixes the natural home of the statement. Consider the prime-factor incidence of the integers: for each n, record which primes divide it (Table 1). Read by rows, the primes are exactly the positions carrying no mark — the survivors of the factorization. Read by columns the table is more telling: a prime p makes its first appearance as a factor at $2p$, its first even multiple, and only from there does it begin to eliminate positions. Consequently the base of primes below a scale P already accounts for every composite up to $2P$: inside $(P, 2P]$ every eliminated position is struck by a prime $< P$, so every survivor there is a new prime. What the observation measures is thus the *reach of deterministic factorization* — how far a fixed base generates the integers before a genuinely new prime is required — and that reach is exactly $2P$. Whether the interval $(P, 2P]$ happens to contain a prime was not the starting point; it is a corollary of the reach, read off afterwards.

Independence and naming. The right endpoint $2P$, the notion of a complete base, and the generative sieve built on them (§2–§6) were arrived at from this observation, independently of the classical postulate and of Erdős's proof; the coincidence of the tightest case with the interval of Bertrand's postulate was recognized only afterwards. We keep the classical name by historical priority, while noting that the statement here is naturally one about *primes* rather than natural numbers: the object is the window $(P_{\max}, P_{\min}P_{\max}]$ anchored at a prime scale, not $(n, 2n]$ anchored at an arbitrary integer, and the two coincide precisely at the tightest member $P_{\min} = 2$ (Corollary 9.3).

n	2	3	5	7	11	13	17	19	23
2									
3									
4	•								
5									
6	•	•							
7									
8	•								
9		•							
10	•		•						
11									
12	•	•							
13									
14	•			•					
15		•	•						
16	•								
17									
18	•	•							
19									
20	•		•						
21		•		•					
22	•				•				
23									
24	•	•							

TABLE 1. Prime-factor incidence for $2 \leq n \leq 24$. A mark in column p records $p \mid n$; the empty rows ($n = 2, 3, 5, 7, 11, 13, 17, 19, 23$) are the primes, the survivors of the factorization. Each prime p first appears as a factor at $2p$, so the primes ≥ 13 carry no mark below $2 \cdot 13 = 26$: the rule at $P_{\max} = 11$ separates the emerged base from the primes not yet required. The base of primes $< P$ thus covers every composite in $(P, 2P]$, and the reach of the fixed base is exactly $2P$ — the endpoint derived in §3.

Throughout, P denotes a prime playing the role of the current scale (the anchor $P_{\max}$); $\mathcal{B}(P) = \{q \text{ prime} : q < P\}$ is the base below it; and, for a finite set S of primes, a position n is *S-void* if no element of S divides n. All Lean identifiers are typeset as `like_this` and refer to the accompanying formalization,[1] in which each may be located by name.

2. THE PRIME BASE AND ITS DETERMINISTIC REACH

Definition 2.1 (Base and void)**.** For a prime P let the *complete base below* P be $\mathcal{B}(P) := \{q \text{ prime} : q < P\}$, and let the *Bertrand window* be $W(P) := (P, 2P] \cap \mathbb{Z}$. A position n is *$\mathcal{B}(P)$-void* if $q \nmid n$ for every $q \in \mathcal{B}(P)$, equivalently if $\gcd(n, \prod_{q<P} q) = 1$.

Formalization. `isVoid`, `isVoid_iff_coprime`, `gps_window`; $\prod_{q<P} q$ is `primorial_below`.

Remark 2.2 (One scale parameter fixes base, window, and width)*.* A single prime P determines all three data: it is the largest prime of the base (the complete set of primes up to P, with none omitted), the left endpoint of the window $W(P) = (P, 2P]$, and its width $|W(P)| = P$. The base and the window are thus both functions of P, not independent parameters. (Whether P itself is counted in the base is immaterial to the covering: its only multiple in the window is $2P$, already divisible by 2; see `anchor_idle_in_own_window`.) The same structure underlies Bertrand's classical window $(n, 2n]$, with n in the role of P: the present statement is phrased on a prime scale rather than an arbitrary integer, and the

[1] Publicly available at `https://github.com/Flaman55/RelMathApps/tree/main/number_theory/structural_bertrand`.

two coincide at the extremal member $P_{\min} = 2$ (§9). The quantity that governs the statement is accordingly the local deficiency of the base on its own window — the length of the longest interval of covered positions in $W(P)$ — which external gap functions only bound from above.

The one arithmetic fact that governs the window is the least-prime-factor bound: a composite that is not too large must reveal a small factor.

Lemma 2.3 (Least prime factor in the window)**.** *Let* $2 < P$ *and* $n \in W(P)$*. If* n *is composite then its least prime factor satisfies* $\mathrm{minFac}(n) \le \sqrt{2P} < P$*; in particular* $\mathrm{minFac}(n) \in \mathcal{B}(P)$*.*

Proof. If $n = ab$ with $1 < a \le b$ then $a \le \sqrt{n}$. For $n \in W(P)$ we have $n \le 2P$, so the least prime factor is $\le \sqrt{2P}$, and $\sqrt{2P} < P$ for $P > 2$. □

Formalization. `least_prime_factor_bound` (`LPF.lean`), `window_composite_minFac_sq_le` (`Rings.lean`).

Lemma 2.3 turns void into primality inside the window: if a position survives the base, it has no factor below P and is not composite, hence prime.

Theorem 2.4 (Determinism of the window)**.** *Let* P *be prime with* $2 < P$ *and* $n \in W(P)$*. Then* n *is* $\mathcal{B}(P)$*-void if and only if* n *is prime.*

Proof. If n is prime and $n > P$ then no $q < P$ divides n, so n is void. Conversely, a void n has no prime factor below P; were it composite, Lemma 2.3 would supply one. Hence n is prime. □

Formalization. `void_in_window_prime`, `void_iff_prime_in_window`, `void_iff_prime_in_deterministic_` the underlying "composite ⇒ covered by an earlier prime" step is `composites_covered_by_prev` and `prime_iff_uncovered_by_prev` (`ZeroForce.lean`).

Two consequences frame the rest of the paper. First, non-emptiness of $W(P)$ as a set of primes is *equivalent* to the existence of a single $\mathcal{B}(P)$-void position; this is the object we will count. Second, the mechanism of Theorem 2.4 does not depend on P: it has the same form at every anchor.

Proposition 2.5 (Scale invariance of the mechanism)**.** *For every prime* P *with* $2 < P$ *and every* $n \in W(P)$*, the equivalence* $(n$ *is* $\mathcal{B}(P)$*-void*$) \Leftrightarrow (n$ *is prime*$)$ *holds verbatim. The rule is invariant under the choice of anchor.*

Formalization. `mechanism_scale_invariant` (`Rings.lean`).

Proposition 2.5 is the precise content of the informal statement that the construction "behaves the same at the start of the number line and at any later point." We return to it in §4: the pair $(W(P), \mathcal{B}(P))$ is a single object viewed at scale P, and moving along $\mathbb{N}$ rescales window and base together.

3. The buildability edge: why $P_{\min}P_{\max}$

The window's right endpoint is not chosen; it is the exact reach of the base. Fix the smallest prime allowed to participate in the sieve, $P_{\min}$ (for Bertrand, $P_{\min} = 2$), and a scale $P_{\max}$.

Theorem 3.1 (Buildability)**.** *Let* $2 \le P_{\min}$ *and let* n *be composite with* $\mathrm{minFac}(n) \ge P_{\min}$ *and* $n \le P_{\min}P_{\max}$*. Then every prime factor of* n *is* $\le P_{\max}$*; that is,* n *is fully built from primes in* $[P_{\min}, P_{\max}]$*.*

Proof. Let r be any prime factor of n and write $n = r\,s$. Since n is composite, $s \ge \mathrm{minFac}(n) \ge P_{\min}$, hence $r = n/s \le P_{\min}P_{\max}/P_{\min} = P_{\max}$. □

Formalization. `rough_composite_smooth` (general $P_{\min}$), `window_composite_smooth` (case $P_{\min} = 2$).

Thus below $P_{\min}P_{\max}$ every composite with least factor $\geq P_{\min}$ is $P_{\max}$-smooth: the base can build it, and a surviving position can only be a new prime. The edge is sharp.

Proposition 3.2 (First unbuildable composite)**.** *The smallest composite with least factor $\geq P_{\min}$ that is* not *$P_{\max}$-smooth is $P_{\min} \cdot \mathrm{nextprime}(P_{\max})$, which exceeds $P_{\min}P_{\max}$. Hence the reach $P_{\min}P_{\max}$ is exact.*

For $P_{\min} = 2$ the self-contained reach is $2P_{\max}$. Two distinct boundaries meet the base here and must not be conflated. The base's *self-containment* ends at $2P$: just beyond it a composite requiring a prime outside the base appears (Proposition 3.2), and $2P$ is the largest interval above P in which no element properly divides another (Theorem 3.4); this is what fixes the window. The sieve's *generative decision* — the equivalence void $\Leftrightarrow$ prime of Theorem 2.4 — reaches much further, throughout the zone (P, P^2), and breaks only at the square of the anchor.

Theorem 3.3 (Generative decision reaches the square scale)**.** *Let P be prime, $2 < P$. Every $\mathcal{B}(P)$-void n with $P < n < P^2$ is prime, and this fails at $n = P^2$: the square P^2 is composite yet $\mathcal{B}(P)$-void. Hence the generative decision of Theorem 2.4 reaches exactly the zone (P, P^2) — the square of the anchor — far beyond the self-contained edge $2P$.*

Proof. If $P < n < P^2$ is $\mathcal{B}(P)$-void then no prime $< P$ divides it, so every prime factor of n is $\geq P$; were n composite it would be a product of two such factors, hence $\geq P^2$, against $n < P^2$. So n is prime. For the failure, P^2 is composite and its only prime factor is $P \notin \mathcal{B}(P)$, so it is $\mathcal{B}(P)$-void; the least prime $q > P$ gives a further composite void $q^2 > P^2$. ☐

Formalization. Positive zone `void_iff_prime_in_deterministic_zone`; the break is witnessed in `determinism_breaks_above` (`Rings.lean`), whose exhibited composite void is q^2.

It is self-containment, not the generative decision, that isolates *why* $2P$ is the window's true boundary: $(P, 2P]$ is exactly the largest interval above P in which no element properly divides another, so elimination can proceed only through the base and never through a survivor.

Theorem 3.4 (Self-containment, sharp at $2P$)**.** *For $m, n \in W(P)$, $m \mid n$ implies $m = n$: no proper divisibility holds inside $W(P)$. This fails immediately above the edge: $P + 1 \in W(P)$ but its proper multiple $2(P + 1) > 2P$ leaves the window.*

Proof. If $m \mid n$ with $m, n \in (P, 2P]$ and $m \neq n$, then $n \geq 2m > 2P$, contradicting $n \leq 2P$. The element $2(P+1)$ is the first proper multiple of an in-window element to exceed $2P$. ☐

Formalization. `window_self_contained_dvd`, `self_containment_breaks_just_above` (`Rings.lean`); `window_self_contained_bound`, `max_self_contained_width` (`SelfContained.lean`).

Self-containment is not a threshold phenomenon that appears only for large bases; it is universal, and a small window is not an exception but merely too narrow to contain such a proper multiple. It is a structural property specific to the tightest window; §4 shows that it is present exactly where the family is most constrained.

4. The generalized window family

Definition 4.1 (Generalized window)**.** For $P_{\min}, P_{\max}$ let $\mathrm{gW}(P_{\min}, P_{\max}) := (P_{\max}, P_{\min}P_{\max}]$ and say it *has a void* if some $n \in \mathrm{gW}(P_{\min}, P_{\max})$ is $\mathcal{B}(P_{\max})$-void.

Formalization. `gwindow`, `gwindowHasVoid` (`Rings.lean`).

Proposition 4.2 (Bertrand is the case $P_{\min} = 2$)**.** $\mathrm{gW}(2, P_{\max}) = W(P_{\max})$, *and* $\mathrm{gW}(2, P_{\max})$ *has a void iff $W(P_{\max})$ contains a prime.*

Formalization. `gwindow_two_eq_window`, `gwindowHasVoid_two_iff`.

The family is monotone in the width parameter: widening the window can only help.

Theorem 4.3 (Monotonicity: tight $\Rightarrow$ wide)**.** *If $P_{\min} \le P'_{\min}$ then $\mathrm{gW}(P_{\min}, P_{\max}) \subseteq \mathrm{gW}(P'_{\min}, P_{\max})$; hence a void for the narrower window is a void for the wider one. In particular every member of the family follows from the tightest member $P_{\min} = 2$.*

Proof. $P_{\min}P_{\max} \le P'_{\min}P_{\max}$ gives the inclusion of half-open intervals with the same left endpoint; a void position and its voidness are preserved. □

Formalization. `gwindow_subset`, `gwindowHasVoid_mono`, `gwindowHasVoid_of_bertrand`.

At the wide end the statement is free: once $P_{\min}P_{\max}$ is large enough, a prime in the window is delivered by elementary means (a Euclid-type witness), with no counting.

Formalization. `gwindowHasVoid_of_wide`.

Two points deserve emphasis. The family is *equivalent* to Bertrand, not strictly stronger: by Theorem 4.3 the narrow case implies the wide cases, so the content of the family is concentrated at $P_{\min} = 2$. Moreover the structure is richest there: self-containment (Theorem 3.4) holds *only* for $P_{\min} = 2$, since any wider window already contains $2(P_{\max} + 1)$, a proper multiple of an in-window element. The tightest window thus carries the most structure — precisely the case that is hardest to establish.

This is the place to state the scale invariance the family embodies. The invariant is not a reflection of coordinates but a dilation: the object is the pair $\big((X, 2X],\ \{q \text{ prime} : q < X\}\big)$, and under $X \mapsto \lambda X$ the window and the base rescale *together*, the ratio 2 being preserved. Freezing the base at one scale while sliding the window to another is not a configuration of the family at all — it mismatches the two dilations. Bertrand's $(n, 2n]$ is simply the unit cell of this scale, $P_{\min} = 2$; motion along $\mathbb{N}$ is the renormalization $n \mapsto \lambda n$ with window and base co-scaling. Proposition 2.5 is the invariance of the local rule; the count that follows inherits it exactly (§6, `coverFiber_eq_scaled`).

5. Truncated arithmetic: the reduced-base sieve

The family of §4 is produced by *truncated arithmetic*: instead of the full base one sieves with the reduced base of primes that can actually reach the window. By Lemma 2.3 the only rings that cover a composite $n \le 2P$ are those with $p \le \sqrt{2P}$; equivalently $p^2 \le 2P$. This selects a truncated primorial.

Definition 5.1 (Truncated primorial)**.** $M_{\mathrm{tr}}(P) := \displaystyle\prod_{\substack{p<P \text{ prime} \\ p^2 \le 2P}} p.$

Formalization. `truncPrimorial`; general version `gtruncPrimorial`.

Theorem 5.2 (Void through the truncated base)**.** *For $2 < P$ and $n \in W(P)$, n is $\mathcal{B}(P)$-void iff $\gcd(n, M_{\mathrm{tr}}(P)) = 1$. Consequently $W(P)$ has a void iff $(P, 2P]$ contains an integer coprime to $M_{\mathrm{tr}}(P)$.*

Proof. Coprimality to the full base clearly implies coprimality to the truncated base. Conversely, if n is coprime to $M_{\mathrm{tr}}(P)$ but divisible by some prime $p < P$, then n is composite, so by Lemma 2.3 $\mathrm{minFac}(n)^2 \le 2P$, and $\mathrm{minFac}(n)$ divides $M_{\mathrm{tr}}(P)$ — contradiction. □

Formalization. `void_iff_coprime_trunc`, `windowHasVoid_iff_trunc_coprime`, `survivor_of_trunc_copr` general `gvoid_iff_coprime_trunc`, `gwindowHasVoid_iff_trunc_coprime`.

The truncation principle makes the reach explicit at every level and exposes the self-similarity. Removing only the prime 2 leaves reach $3n$; removing 2 and 3 leaves $5n$; in general the smallest omitted prime dictates the reach. The buildability map is therefore layered: the class of positions with least factor p is buildable up to $p\,P_{\max}$, and the same phenomenon recurs at each level with Bertrand as the base case $P_{\min} = 2$.

Formalization. Reduced-base sieve and its non-closure: `exists_uncovered_of_card_lt`, `truncated_not_sieve_closed`, `card_truncated_window` (`Truncated.lean`).

6. Reduction to a single window-count atom

We now count. Let $\operatorname{cov}(P)$ be the number of composite (covered) positions in $W(P)$; since $|W(P)| = P$, non-emptiness is a strict inequality.

Definition 6.1 (Covered count)**.** $\operatorname{cov}(P) := \#\{n \in W(P) : n \text{ is not } \mathcal{B}(P)\text{-void}\}$.

Theorem 6.2 (Reduction to an atom)**.** *$W(P)$ has a void iff $\operatorname{cov}(P) < P$. Hence Bertrand at scale P is equivalent to the single positivity statement $P - \operatorname{cov}(P) > 0$.*

Formalization. `coveredCount`, `gps_window_card`, `windowHasVoid_iff_coveredCount_lt`, `coveredCount_lt_imp_nonempty`.

A structural preliminary shows the reduction is not vacuous: the largest ring $P_{\max}$ exerts no force inside its own window (its first multiple $2P_{\max}$ is the lone edge point), and the previous base carries a weight $w = P_{\max} \cdot \varphi(M)/M \geq 1$. The weight being ≥ 1 is the necessary condition for a void; the atom is the sufficiency.

Formalization. `zero_effective_force` (`ZeroForce.lean`); weight $w \geq 1$: `WeightGeOne`, `weight_base`, `weight_step`, `structural_weight_ge_one`, `prev_base_cannot_cover_window` (`Weight.lean`); bridge `weight_density_bridge`; anchor idle `anchor_idle_in_own_window`.

The covered count decomposes disjointly by least prime factor, and each fiber is, by an exact bijection, the *same* coprimality count one scale down — the recursion engine promised by scale invariance.

Theorem 6.3 (Disjoint minFac decomposition and self-similar fibers)**.** $\operatorname{cov}(P) = \sum_p A_p(P)$, *where $A_p(P)$ counts the $n \in W(P)$ with $\operatorname{minFac}(n) = p$, the sum ranging over primes with $p^2 \leq 2P$. Moreover*

$$A_p(P) = \#\Big\{m \in (\tfrac{P}{p}, \tfrac{2P}{p}] : \gcd\big(m, \textstyle\prod_{r<p} r\big) = 1\Big\},$$

the same window-coprimality problem at scale P/p.

Formalization. `coveredCount_eq_sum_minFac_fiber`, `coveredCount_eq_sum_truncated`, `coverFiber_eq_scaled`; the dominant ring $p = 2$ is split off exactly in `coveredCount_split_two`, `windowHasVoid_iff_oddSum_lt`.

Two exact handles on $\operatorname{cov}(P)$ follow. A union bound with multiplicity gives $\operatorname{cov}(P) \leq \sigma(P) := \sum_p(\text{hits of } p)$; the gap between the distinct count $\operatorname{cov}(P)$ and $\sigma(P)$ is exactly the overlap (redundancy) between rings, so $\sigma(P) > P$ does not obstruct $\operatorname{cov}(P) < P$. The full, signed identity is inclusion–exclusion (Legendre).

Theorem 6.4 (Interference identity)**.** *For a finite set S of primes and $a \leq b$,*

$$\#\Big\{n \in (a, b] : \gcd\big(n, \textstyle\prod_{q\in S} q\big) = 1\Big\} = \textstyle\sum_{T\subseteq S}(-1)^{|T|}\Big(\big\lfloor \frac{b}{\prod_{q\in T} q}\big\rfloor - \big\lfloor \frac{a}{\prod_{q\in T} q}\big\rfloor\Big).$$

The order-by-order layers are its partial sums; the whole is an exact identity.

Formalization. `coveredCount_le_sigma`, `interference_step`, `interference_formula` (`Rings.lean`).

Entire regimes of the atom close with no analysis. In the *sparse* regime a single union bound produces a survivor. For small anchors a sufficient condition settles the window: only the *active-covering rings* $p^2 \leq 2P$ can cover a composite in $W(P)$ (Lemma 2.3), so, writing $M_{\mathrm{tr}}(P) = \prod_{p^2\leq 2P} p$, if every run of P consecutive integers meets a number coprime to $M_{\mathrm{tr}}(P)$ — i.e. the coprime-gap $g(M_{\mathrm{tr}}(P)) \leq P$ — then $W(P)$ contains a coprime, hence a void.

Theorem 6.5 (Small-anchor closure). *If $g(M_{\mathrm{tr}}(P)) \le P$ then $W(P)$ has a void. Concretely $g(30) = 6$, $g(210) = 10$, $g(2310) = 14$ close all anchors $P \le 83$ unconditionally.*

A caveat on this bound. Here $g(M_{\mathrm{tr}}(P))$ is the global coprime-gap of a *fixed* modulus — the product of the active-covering rings, a subcollection that coincides with a smaller, foreign base, not the complete base of the object. It only *upper-bounds* the object's own quantity, the local run of covered positions inside $W(P)$; the inequality $g \le P$ is therefore a loose sufficient proxy, not the governing structure of the object, and it ceases to be available at large P, where the residual dense regime is settled by the object below, not by any gap function.

Formalization. `windowHasVoid_of_jacobsthal`, `windowHasVoid_of_trunc_gap`, `jacobsthal_thirty`, `jacobsthal_210`, `jacobsthal_2310`, and `windowHasVoid_3` ... `windowHasVoid_83` (`Rings.lean`).

What remains is the *dense* regime at large P. Here the structural mechanism has a definite reach and a definite limit, and it is worth stating both precisely. At every *concrete* scale P the construction decides the window outright: the sieve is finite and deterministic, so each finite step is verified and no counterexample can arise — the statement "$W(P)$ has a void" is decidable at every fixed P. What the generative process cannot deliver is the *uniform* statement over all scales at once: to reach "for every P" by generation would mean carrying the construction to infinity, an infinite process that no finite computation realizes. The passage from "true at every finite step" to "true in the infinite" is therefore not computational but *existential.* We do not claim a generative closure of this existential step, and none is known. It is precisely this closure of the infinite tail — the part beyond any finite computation — that the central binomial coefficient performs, and the next section explains why it is the appropriate object for this step.

Formalization. Regime dispatch: `structural_sieve_survivor`; the residual dense case `dense_sieve_survivor` (`GPS_StateMachine.lean`).

7. The central binomial coefficient and the window's prime content

The dense regime is closed existentially, and the central binomial coefficient is the instrument of that closure. It is invoked for one purpose only: to settle the infinite tail that no finite generation can reach — a uniform argument in place of an unrealizable infinite process, not a repair of the finite mechanism, which needs none, since that mechanism already decides every concrete scale. The coefficient need not be present for any fixed window; it is present because the proof must quantify over all windows at once, and it supplies closure over that range.

That $\binom{2n}{n}$ is the appropriate object here is itself structural. It is an entry of Newton's binomial expansion: $4^n = (1+1)^{2n}$ is the sum of the $2n+1$ entries of the $2n$-th Pascal row, of which the central one is largest, and the window's new sources are recorded in its prime factorization by a purity law internal to the construction.

Theorem 7.1 ($S1$ purity law). *A new source $q \in (n, 2n]$ divides $\binom{2n}{n}$ exactly once: its first multiple $2q$ already exceeds $2n$, so q contributes no further factor to $(2n)!$, while $q > n$ keeps it out of $(n!)^2$. Consequently the product of all new sources of the window divides $\binom{2n}{n}$:*

$$\prod_{\substack{q \in (n,2n] \\ q \text{ prime}}} q \;\Big|\; \binom{2n}{n}, \qquad \textit{hence} \qquad \prod_{\substack{q \in (n,2n] \\ q \text{ prime}}} q \;\le\; \binom{2n}{n}.$$

Formalization. `window_primes_prod_dvd_centralBinom` (`Rings.lean`), `window_primes_prod_dvd_central`
`window_primes_prod_le_centralBinom` (`Newton.lean`); the divisibility rests on `prod_primes_dvd` and `Nat.Prime.dvd_choose_add`. The purity is the same self-containment property as `window_self_contained_dvd` / `window_composite_smooth` — the first proper multiple of

an in-window source already exceeds $2P_{\max}$ — with the same role of 2 and the same extremal self-contained case $P_{\min} = 2$.

The choice of object is thus forced by the structure, not imported from history: the window's new sources are recorded in the prime factorization of $\binom{2n}{n}$. Its role in the proof is existential — it closes what cannot be generated — while its identity is structural. Two bounds close on it. The lower bound is pure combinatorics of the Pascal row.

Theorem 7.2 (Lower bound)**.** *$4^n \le (2n+1)\binom{2n}{n}$, proved directly from $4^n = \sum_{i=0}^{2n} \binom{2n}{i}$ with each term $\le \binom{2n}{n}$.*

Formalization. `four_pow_le_newton` (`Newton.lean`).

The upper bound reads the same factorization in the contrapositive: an empty window would force $\binom{2n}{n}$ to be built entirely from old sources.

Theorem 7.3 (Empty window $\Rightarrow$ old sources only)**.** *If $(n, 2n]$ contains no prime, then every prime factor of $\binom{2n}{n}$ is $\le n$.*

Proof. Any prime factor of $\binom{2n}{n}$ divides $(2n)!$, hence is $\le 2n$; absent new sources in $(n, 2n]$ it is $\le n$. □

Formalization. `centralBinom_prime_factor_le` (`Newton.lean`).

8. Closing the extremal window: the self-contained certificate

Only the extremal window $P_{\min} = 2$ needs closing, and only in the dense regime. We split at $n = 512$.

For $2 < n < 512$ a kernel-checked decision procedure exhibits a prime in $(n, 2n]$ directly: the range is split into finitely many blocks, and on each a witness prime is exhibited and verified by direct reduction inside Lean's trusted kernel — no external lemma, and no appeal to compiled evaluation.

Formalization. `small_window_oracle`, by `decide` (the range split via `interval_cases` into blocks the kernel evaluator can handle in one call).

For $n \ge 512$ the two bounds are combined. Suppose $(n, 2n]$ has no prime. Theorem 7.3 bounds $\binom{2n}{n}$ by its old-source content, at most $(2n)^{\sqrt{2n}}\, 4^{2n/3}$; combined with the lower bound and the size threshold one obtains $4^n < 4^n$, a contradiction.

Theorem 8.1 (Quantitative kernel)**.** *For $2 < n$, if no prime lies in $(n, 2n]$ then a contradiction follows. Hence every window $(n, 2n]$ with $n > 1$ contains a prime.*

Proof sketch. For $n < 512$ use the oracle. For $n \ge 512$, chain the lower bound $4^n < n\binom{2n}{n}$ (`Nat.four_pow_lt_mul_centralBinom`), the old-source upper bound $\binom{2n}{n} \le (2n)^{\sqrt{2n}} 4^{2n/3}$ (`window_centralBinom_le`, reproved in-project from Legendre/Kummer and primorial primitives), and the prime-free size comparison $n\,(2n)^{\sqrt{2n}} 4^{2n/3} \le 4^n$ (`threshold_inequality`) to reach $4^n < 4^n$. □

Formalization. `binomial_contradiction` (`BinomialCertificate.lean`). The upper bound (`window_centralBinom_le`, `BinomialBound.lean`) is reproved from `Choose.Factorization` and `Primorial`; the size threshold (`threshold_inequality`, `Threshold.lean`) from real convexity. Neither imports `Mathlib.NumberTheory.Bertrand`, so the main theorem's import closure is free of it. The certificate is modular (`WindowCertificate`, `Certificate.lean`): any estimate of the same strength — a Chebyshev-type bound, or Nair's $\operatorname{lcm}(1, \dots, 2n)$ argument [5] — plugs into the identical slot; a second instance, `erdos_certificate`, does so through Mathlib's inequalities.

Non-circularity is by inspection: Mathlib's `Nat.bertrand` and `exists_prime_lt_and_le_two_mul` are never used, and `Mathlib.NumberTheory.Bertrand` is not in the import closure of the main theorem; it appears only in `Erdos.lean`, which provides the instance-A comparison and is not imported by the main theorem.

9. Main theorem and recovery of the classical case

Assembling the reduction (§6), the factorization identity (§7) and the certificate (§8) closes the atom at every scale, hence produces a prime in every window.

Theorem 9.1 (Window non-emptiness). *For every prime P with $2 < P$, the window $(P, 2P]$ contains a prime.*

Formalization. `structural_sieve_survivor` $\to$ `gps_window_nonempty` $\to$ `prime_in_window` (`GPS_StateMachine.lean`).

Theorem 9.2 (Generalized family). *For every prime $P_{\max}$ and every $P_{\min} \geq 2$, the window $(P_{\max}, P_{\min}P_{\max}]$ contains a prime.*

Proof. By Theorem 4.3 it suffices to treat $P_{\min} = 2$, which is Theorem 9.1. □

Formalization. `gwindowHasVoid_of_bertrand`.

Corollary 9.3 (Classical Bertrand–Chebyshev). *For every integer $N > 1$ there is a prime p with $N < p \leq 2N$.*

Proof. Let P be the largest prime $\leq N$. If $P = 2$ then $N = 2$ and $p = 3$ works. Otherwise $2 < P$; Theorem 9.1 gives a prime $q \in (P, 2P]$, and maximality of P forces $q > N$, while $P \leq N$ gives $q \leq 2N$. □

Formalization. `exists_largest_prime_le`, `bertrand_chebyshev` (`Main.lean`); the base-consistent structural form is `structural_bertrand_chebyshev`.

The constant 2 is not arbitrary: it is the smallest prime, hence the smallest admissible $P_{\min}$, hence the buildability edge (§3). Bertrand's window is thus the extremal member of the family — the case $P_{\min} = 2$, on which all wider members depend and at which the self-containment structure is present.

10. Remarks and provenance

Machine verification. The development is checked in Lean 4 with Mathlib (toolchain pinned in the repository); there is no `sorry`, and the main theorem depends on no axiom beyond the three standard ones (`propext`, `Classical.choice`, `Quot.sound`), verified directly by `#print axioms bertrand_chebyshev` — in particular `Lean.ofReduceBool` (the axiom `native_decide` introduces) does not appear anywhere in its import closure. The small-window oracle for $n < 512$, the base anchors $P \leq 83$, and the Jacobsthal gap values are all discharged by `decide`, fully reduced and checked by Lean's own kernel. The two widest of these — the $n < 512$ search and the $g(2310)$ gap check — exceed the kernel's evaluator as a single call and are each split into blocks small enough for it to handle, rather than falling back to compiled evaluation. (A separate, Mathlib-based comparison instance in `Erdos.lean` does use `native_decide`; it lies outside the main theorem's import closure and does not affect this.) A non-circularity audit (a search over uses) confirms that Mathlib's own Bertrand theorem is not invoked.

Provenance. The development is neither independent of Erdős nor Erdős in disguise. The structural reduction is independent: it derives the window and its edge from the base (§2–§3), assembles the scale-invariant family (§4–§5), and localizes the theorem to one exact-count atom whose small, sparse and gap regimes close with no counting (§6). The atom is settled on the central binomial coefficient, whose prime factorization records the window's content via the $S1$ purity law (§7) and would be reached here regardless of the historical record. Erdős reached the same object in 1932 by a different route: a convergence on one mathematical object, carrying historical priority of use but no exclusivity over it. In the Lean development the two bounds are reproved in-project, so the main theorem's import closure contains no Mathlib Bertrand theorem; the certificate is modular (§8), a second, Mathlib-based instance being retained, outside that import closure, for comparison.

Relation to classical proofs. The successive treatments of the postulate answered different questions and, accordingly, studied different objects, while converging on one theorem. Bertrand [1] posed the question of existence, read from tables of the integers; Chebyshev [2] answered it through the analytic size of ϑ and ψ; Erdős [4] and Nair [5] through the elementary size of $\binom{2n}{n}$ and of $\operatorname{lcm}(1, \ldots, 2n)$. The present treatment asks a fourth question — the deterministic reach of a generating base — and takes the base, not the window, as its primary object; the window and its endpoint are then derived, and the existence of a prime is a corollary of the reach rather than the premise. At the single quantitative point all four are of the same strength, and any of the elementary certificates plugs into the same slot; what the structural layer adds is the reduction that makes one such certificate sufficient, together with the derivation of the window itself from the base.

Further directions. The finite mechanism decides every concrete scale; what a certificate supplies is the existential closure of the infinite tail. Whether that closure can be effected without a size inequality is open; every method known to the author is quantitative, of the same Chebyshev strength.

Competing Interests

The author declares no competing interests.

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